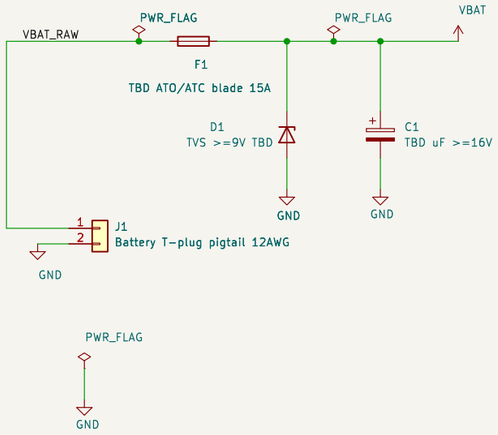
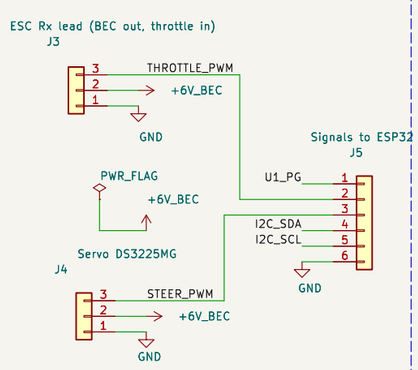
A Battery input (2S LiPo, 6.0–8.4 V)



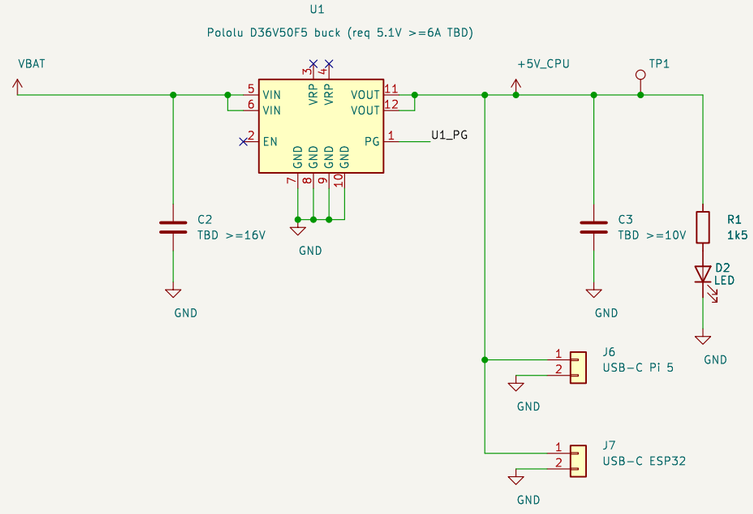
ESC power does NOT pass through this board:
battery -> ESC on its own 12 AWG leads.
J1/F1 carry only the board loads
(about 8.4 A worst case at 6.4 V).

B ESC signals and servo



ESC: up to 20 A motor current (THW–1060: 60 A cont.)
from the battery on its own 12 AWG leads, OFF–BOARD.
Only its 3–wire receiver lead lands here (J3).
J5 (JST–XH 6–pin) to ESP32: 1 U1_PG (GPIO TBD),
2 THROTTLE_PWM GPIO5, 3 STEER_PWM GPIO4,
4 I2C_SDA GPIO8, 5 I2C_SCL GPIO9, 6 GND. 3.3 V logic.

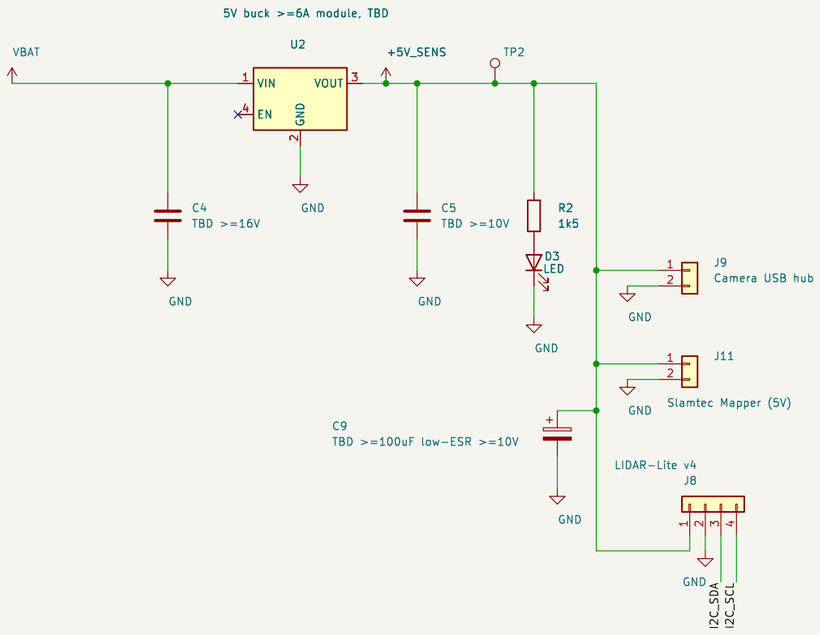
C Compute 5 V (+5V_CPU: Pi 5 + ESP32 only)



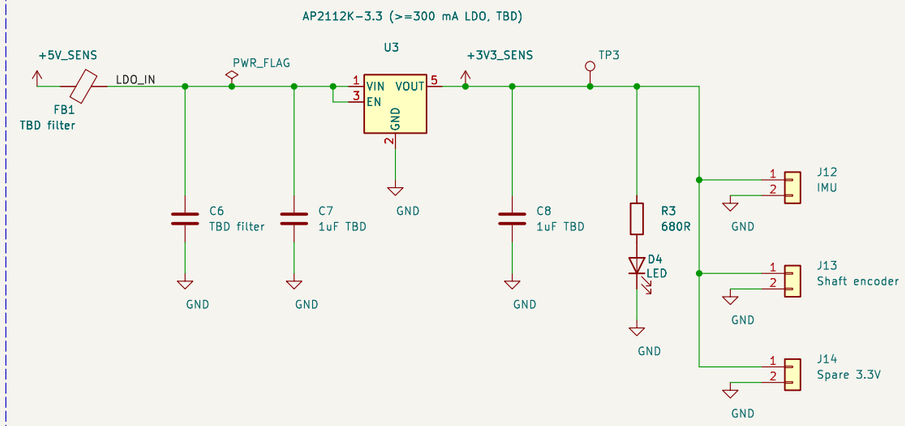
Notes

1. Battery: 2S LiPo, 6.0–8.4 V (ESC LVC cuts at 6.0 V).
   T–plug pigtail (12 AWG bare ends) soldered to J1 pads;
   pad size TBD (footprint is the 2.5 mm2 placeholder).
2. ESC is powered OFF–BOARD: battery -> ESC on its own
   12 AWG leads (up to 20 A). This board carries only its
   own loads (about 8.4 A worst case at 6.4 V, 9.0 A at 6.0 V,
   9.3 A briefly at Mapper startup), so F1 is sized for that,
   not for the motor.
3. F1 is the single replaceable main fuse (standard ATO/ATC
   blade in a Littelfuse FLR 178.6165 PCB holder)
   protecting compute and both sensor rails;
   no branch fuses. 15 A: a 10 A fuse would run at
   84–93 % of its rating.
   Trade–off: a sensor–branch short opens F1 and drops
   compute too (U1/U2 limit their own output current).
   All protection values TBD.
4. Single common GND; star point at J1 in layout.
5. +6V_BEC is ESC–supplied (servo only), never joined
   to a 5 V rail.
6. Signals are 3.3 V logic or lower.
7. U1 Pololu D36V50F5: all 12 pins drawn (2x VIN, 2x VOUT,
   4x GND, 2x VRP, EN, PG). VRP unused (sensor buck fed
   from VBAT). EN left unconnected: the module's
   internal pull–up keeps it on (ESP32 EN control
   archived). PG: open–drain, to ESP32 via J5 (ESP32
   INPUT_PULLUP, non–strapping GPIO TBD).
   J5 (JST–XH 6–pin) to ESP32: 1 U1_PG (GPIO TBD),
   2 THROTTLE_PWM GPIO5, 3 STEER_PWM GPIO4,
   4 I2C_SDA GPIO8, 5 I2C_SCL GPIO9, 6 GND.
8. U2 EN per module datasheet (TBD); U2 footprint TBD.
   U2 must be >=6 A: +5V_SENS is 4.6 A worst case running,
   4.9 A at Mapper startup (proposal Pololu D24V60F5,
   5 V 6 A; check its dropout near the 6.0 V cutoff, TBD).
9. Rails are project power symbols (rc_power lib):
   pin name = Value = net name.
10. PWR_FLAG = ERC marker only (no part, not on the PCB):
    tells ERC where power enters VBAT_RAW, GND, VBAT,
    LDO_IN, +6V_BEC.
11. Net classes (kicad_pro): HighCurrent = *VBAT*, GND;
    Power5A = +5V_CPU, +5V_SENS;
    Power3A = *+6V_BEC.
12. J6/J7: 2–wire USB–C pigtails (20 AWG, 30 cm, about
    0.10 V drop at 5 A; trim to 10–15 cm) into the Pi 5
    and ESP32 USB–C inputs, keeping their own input
    protection. Pi 5 gets no USB–PD 5 A offer (USB ports
    limited to 600 mA unless set in config.txt).
13. J8 Garmin LIDAR–Lite v4: 1 VIN red +5V_SENS,
    2 GND black, 3 SDA white, 4 SCL green; I2C addr
    0x62; no external pull–ups (13k to 3.3 V inside
    the sensor); 3.3 V logic.
14. J11 Slamtec Mapper M2M1 (5 V USB–A socket): harness
    J11 XH -> 22 AWG 3 A USB–A male pigtail, pin 1 +5V_SENS
    red/VBUS, pin 2 GND black. Supply 4.8–12 V, 0.75 A typ,
    1.3 A max running, 1.4–1.6 A startup for a few seconds;
    ripple <=50 mV (20 mV typ). Never from Pi USB or straight
    from the pack. C9 (bulk, TBD >=100 uF low–ESR) sits at
    J11 to carry the startup surge and hold ripple down.
15. Connectors: JST–XH (2.5 mm) is the convention for
    sensors and low–power parts; exceptions J1 (battery
    T–plug pigtail pads) and J6/J7 (USB–C pigtail
    screw terminals, 5 A).
16. Spec: power_board_plan.md; meeting summary:
    design_overview.md; parts: bom.md; layout/fab:
    fab_handoff.md. TBD values are unconfirmed.

D Sensor 5 V (+5V_SENS: heavy sensors + 3.3 V LDO)



E Sensor 3.3 V (+3V3_SENS: LDO from +5V_SENS)



ESP32 3V3 pin must not power these sensors (ESP32 powers nothing).



Meeting summary: docs/schematics/design_overview.md; project KiCad version 10.0
Spec: docs/schematics/power_board_plan.md
Draft for review, values marked TBD are unconfirmed
gvo27/racecar

Sheet: /
File: rc_power_board.kicad_sch

Title: RC Car Power Distribution Board

Size: A3	Date: 2026–10–30	**Rev: A (draft)**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 1/1